

Answer Key — Biology Grade 11

Sub-Strand 1.3: Taxonomy II • Lesson 6: Classes of Kingdom Animalia

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7
D	A	B	D	C	C	A

USE AFTER: PREDICT PHASE — Predict: Think-Pair-Share on which features could split the chicken, lizard and goat into smaller groups

Q1. A goat, a chicken and a lizard are all in Phylum Chordata. Which observable feature can be used to place the chicken in a different class from the goat?

- A. The chicken and the goat are both used as food
- B. The chicken is smaller than the goat
- C. The chicken is usually white or brown in colour
- D. The chicken has feathers, while the goat has hair**

Correct answer: D. Classes are separated by structural features such as body covering. Feathers belong to birds (Aves) and hair belongs to mammals (Mammalia). Size, colour and use as food are the learners' own groupings and are not shared, reliable features.

USE AFTER: OBSERVE PHASE — Jigsaw Part 1: Arthropoda classes expert group (dead mosquito and butterfly, spider picture)

Q2. A student examines a dead mosquito and a spider. Which statement correctly describes the mosquito and its class?

- A. It has three pairs of legs and a body in three parts, so it belongs to Insecta**
- B. It has three pairs of legs and feathers, so it belongs to Aves
- C. It has four pairs of legs and belongs to Arachnida
- D. It has no legs and belongs to Crustacea

Correct answer: A. Insects such as the mosquito have six legs (three pairs), antennae and three body parts. Spiders have eight legs (four pairs) and belong to Arachnida. Counting legs corrects the error that insects have 8 legs.

USE AFTER: EXPLAIN PHASE — Step 2: CER response 'Is a bat a bird or a mammal?'

Q3. A bat can fly. It has hair on its body, feeds its young on milk and gives birth to live young. To which class does a bat belong?

- A. Insecta, because it has wings
- B. Mammalia, because of its hair, milk and live young**
- C. Reptilia, because it has a dry body covering
- D. Aves, because it can fly

Correct answer: B. Class is decided by features such as body covering and reproduction, not by ability to fly. Hair, milk and live young show that a bat is a mammal. Birds are defined by feathers.

USE AFTER: EXPLAIN PHASE — Step 1: Dichotomous-key challenge, groups swap and test keys

Q4. A group writes this key: 1a. Has feathers – Aves; 1b. No feathers – go to 2. 2a. Has hair – Mammalia; 2b. No hair – go to 3. 3a. Has dry, scaly skin – Reptilia; 3b. Has moist skin – Amphibia. Which animal follows 1b, then 2b, then 3a?

- A. A frog
- B. A goat
- C. A chicken
- D. A lizard**

Correct answer: D. A lizard has no feathers, no hair and has dry, scaly skin, so it ends at Reptilia. A chicken stops at 1a, a goat at 2a, and a frog goes to 3b.

USE AFTER: EXPLAIN PHASE — Exit check: 'Name one feature that separates reptiles from amphibians'

Q5. Which feature separates reptiles from amphibians?

- A. Amphibians have hair, but reptiles do not
- B. Reptiles have feathers, but amphibians do not
- C. Reptiles have dry, scaly skin, but amphibians have moist skin without scales**
- D. Reptiles have gills as adults, but amphibians have lungs

Correct answer: C. Reptiles such as the lizard have dry, scaly skin. Amphibians such as frogs have moist skin without scales. Feathers belong to birds and hair belongs to mammals.

USE AFTER: DQB PHASE — Driving Question Board: sticky notes on what we learned about classes of animals

Q6. Which statement best describes how phyla and classes are related in classifying animals?

- A. Classes are made by grouping animals by size and colour
- B. Classes are larger groups that contain several phyla
- C. A phylum is a broad group that is divided into classes using more specific characteristics**
- D. Phyla and classes are the same thing and use the same features

Correct answer: C. A phylum such as Chordata is broad, so it is divided into classes (Pisces, Amphibia, Reptilia, Aves, Mammalia) using features like body covering, limbs, respiration and reproduction. This is the new idea added to the Driving Question Board.

USE AFTER: MODEL BUILDING PHASE — Revising the class classification model on manila paper: placing the phenomenon animals on branches

Q7. When updating a classification model, a group must place a snail. Where does it belong?

- A. In neither Chordata nor Arthropoda, because it belongs to another phylum**
- B. In Phylum Arthropoda, because it has a hard shell
- C. In Phylum Chordata, class Reptilia, because it moves slowly
- D. In Class Insecta, because it is a small animal

Correct answer: A. A snail is an animal but is not a chordate or an arthropod. It belongs to a different phylum, so it is marked separately on the model. A shell is not the same as an arthropod's jointed legs and exoskeleton.